

ChE 382 – Chemical Engineering Laboratory II- Spring 2017

Student Learning Outcomes 1, 3 (written and oral), 4, 5, 6, 7

Rubric for assessment for Pre-lab Report, Final Lab Report, Team Skills Assessment, SACHE Safety Tutorial; Dust Explosion, Safety Quiz: Industrial Hazards

RESULTS BY PERFORMANCE INDICATOR: Prelab and Final Lab Reports											
Rubric item	1	2	3	4	5	6	7	8	9	10	11
Mapped SLO	1,3	3,6	3,4,7	3	1,3,6	1,3,6	3,6	3,7	1,3,6	1,3,6	1,3,6
Average	2.58	3.00	2.82	2.66	2.64	2.42	2.13	2.24	2.82	3.00	3.00
Std Dev	0.50	0.00	0.39	0.48	0.48	0.50	0.34	0.43	0.39	0.00	0.00

RESULTS BY PERFORMANCE INDICATOR: Team Assessment			
Rubric item	12	13	14
Mapped SLO	5	5	5
Average	2.96	2.96	2.93
Std Dev	0.21	0.21	0.26

RESULTS BY PERFORMANCE INDICATOR: Safety		
Rubric item	15	16
Mapped SLO	4	4
Average	3.00	2.93
Std Dev	0.00	0.31

RESULTS BY STUDENT OUTCOME ALL ASSESSMENTS							
	1	3	3	4	5	6	7
		Written	Oral				
Average Score	2.74	2.66	No Data	2.92	2.95	2.72	2.53
Standard dev.	0.24	0.30	No Data	0.09	0.02	0.34	0.41

Average scores showed most students were very strong in experimental preparation, discussion of results, lab and industrial safety, sample calculations, data tabulation and error analysis. A large majority were satisfactory in analyzing the experimental data and drawing conclusions. Most students were also strong in teamwork/team skills.

Assessment Rubric for Lab reports (pre-lab and final lab report sections combined)

Report Section	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1.Introduction	SLO 1, 3	Main aspects of the lab experiments are not addressed; no experimental objective, experimental description or discussion of industrial applications, poor grammar and usage	Several main aspects of lab experiment not addressed or not stated in a clear and concise manner.	Main aspects of lab experiment addressed and clearly stated; minor grammar and usage issues.
2. Experimental: Apparatus, Materials & Supplies, Procedure	SLO 3, 6	Incomplete process diagram, missing materials/supplies list, incomplete or vague experimental procedure	Adequate process diagram, material/ supply list, experimental procedure; several problems or mistakes	Complete process diagram, material/ supply list, experimental procedure with minor problems
3. Project Safety Evaluation	SLO 3, 4, 7	Missing or lacking an understanding of major/ minor safety hazards and safe operating conditions	Safety understanding, missing some major safety hazards	Safety understanding and important safety hazards, missing some minor hazards
4. Executive Summary	SLO 3	Illegible or incoherent summary of conclusions, poor grammar and formatting	A few problems with summary of conclusions, legibility or neatness	Minor problems with summary of conclusions, grammar, formatting, neatness
5. Results	SLO 1, 3, 6	Drawings, graphs and tables unacceptable, confusing, or missing	Several deficiencies in drawings, graphs and tables. Figures and tables not referred to in text.	Only a few deficiencies with drawings, graphs and tables
6. Discussion	SLO 1, 3, 6	Main points and trends from results are missing or not discussed clearly; lack of understanding of the engineering reasoning behind the analysis	Several main points and trends are missing or not discussed clearly; some understanding of engineering reasoning behind analysis	Only a few points or trends are missing or not discussed clearly; a substantial understanding of the engineering reasoning behind analysis
7. Conclusions	SLO 3, 6	No significant conclusions drawn, Limitations of results and future work not discussed.	Cursory conclusions drawn from data and calculations. Some discussion of limitation of results and future work.	Valid conclusions drawn from data and calculations. Future work addressed in relation to limitations or uncertainties.
8. References	SLO 3, 7	Seeks no extra information other than what is provided by instructor	Seeks information from a few sources, mainly from the textbook or instructor	Seeks information on problems from multiple sources

9. Data Tabulation	SLO 1, 3, 6	Experimental data tables and graphs unacceptable or missing	Several deficiencies with data tables and graphs	Only a few deficiencies in data tables and graphs
10. Sample Calculations	SLO 1, 3, 6	Calculations are incorrect, presented in a confusing way and do not link with specific experimental results	Some calculations are incorrect or presented in a confusing way	One or two calculations are incorrect or presented in a confusing way
11. Error Analysis	SLO 1, 3, 6	Cursory and qualitative description of errors	Qualitative analysis of error with some calculations	Quantitative and qualitative error analysis, missing some minor calculations

Assessment Grading for Lab reports (pre-lab and final lab report sections combined)

Report Section	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Introduction	SLO 1, 3	0-2	3	4-5
2. Experimental: Apparatus, Materials & Supplies, Procedure	SLO 3, 6	0-11	12-15	16-20
3. Project Safety Evaluation	SLO 3, 4, 7	0-5	6-7	8-10
4. Executive Summary	SLO 3	0-5	6-7	8-10
5. Results	SLO 1, 3, 6	0-8	9-11	12-15
6. Discussion	SLO 1, 3, 6	0-8	9-11	12-15
7. Conclusions	SLO 3, 6	0-6	7-8	9-10
8. References	SLO 3, 7	0-2	3	4-5
9. Data Tabulation	SLO 1, 3, 6	0-8	9-11	12-15
10. Sample Calculations	SLO 1, 3, 6	0-8	9-11	12-15
11. Error Analysis	SLO 1, 3, 6	0-8	9-11	12-15

Assessment Rubric for Team Member Skills

Performance Indicator (PI)	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
12. Quality Work	SLO 5	Has a poor grasp of chemical engineering principles and poor writing skills	Has some knowledge of chemical engineering principles and writing skills	Has good knowledge of chemical engineering principles and writing skills
13. Teamwork	SLO 5	Hides in the background; only participates if strongly encouraged Routinely fails to prepare any plans, does a minimal amount of task work	Occasionally plans and carries out lab tasks. Occasionally takes charge when not in the position to lead	Demonstrates the ability to assume a designated role in the group, plan tasks, carry out tasks and contributes to success of team
14. On-time tasks	SLO 5	Consistently hands in task late to the team	Occasionally is late in handing in tasks	Hands in tasks (lab report sections) on time to team

Team Member Assessment Evaluation Form:

CHE 381 TEAM MEMBER ASSESSMENT SHEET

Submitted by: _____ Team # _____ Date: _____

The team member review form is given below and shall be submitted to Professor Zdunek by email on or before 5PM Monday December 7th.

Please be fair and honest in your assessment. I will take the team grading into consideration when preparing the Chemical Engineering Laboratory grades-

- A lack of positive, meaningful team participation can affect your individual grade.
- If you do not submit a team member review form on-time, your grade will also be impacted.

Thank you!

TEAM MEMBER YOU ARE ASSESSING: _____.

1 = never, 5 = always

	1	2	3	4	5
Did he/she submit quality, complete work for assigned actions					
Did he/she participate in a cooperative and respectful manner					
Did he/she meet promised dates for assigned actions					

TEAM MEMBER YOU ARE ASSESSING: _____.

1 = never, 5 = always

	1	2	3	4	5
Did he/she submit quality, complete work for assigned actions					
Did he/she participate in a cooperative and respectful manner					
Did he/she meet promised dates for assigned actions					

Assessment Rubric for Safety Assignments

Performance Indicator (PI)	SLO's	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
15. Understanding industrial safety and practices	SLO 4	Missing or lacking an understanding of industrial safety practices	Has some understanding of industrial hazards and safety practices	Has a good understanding of industrial hazards and safety practices Example: Passed a safety tutorial (SACHE)
16. Understanding of lab safety and safety evaluations	SLO 4	Missing or lacking an understanding of major/ minor safety hazards and safe operating conditions	Missing some major/minor aspects of laboratory safety practices	Satisfactory understanding of laboratory safety covered. Example: Understands a laboratory Chemical Hygiene Plan

Safety Assignment: SACHE Tutorial "Dust Explosion"

SAFETY ASSIGNMENT II: AICHE (American Institute of Chemical Engineers) has created various safety tutorials covering process safety, risk assessment and chemical reactivity hazards.

Complete the tutorial titled "**Dust Explosion**" in the SACHE Certificate Program.

DOWNLOAD PROCEDURE:

Go to : <http://www.aiche.org/academy/courses/ela904/sache-certificate-program-chemical-reactivity-hazards>

Click "Checkout"

Login with your student AICHE username and password; if you logged in with your student member number the price for the tutorial should change to \$0.00

Complete Checkout by clicking on "Process my order" and download the tutorial; there will be many files and subdirectories that will be downloaded.

The course is distributed as a zipped file. Once you 'purchase' it, you can find it when you are logged into the AICHE website under the 'academy' link, and then further in "My Stuff". It should be listed there; you can click on it and get into the main menu for the tutorial. From there you can download the file to your computer and then unzip it to extract the files, and then start the course by opening the "index.htm" file.

RUNNING THE TUTORIAL:

To access the tutorial double-click on the 'Index.htm' file which will start the tutorial.

Follow the directions and answer the sets of questions to move to the next section.

When you successfully complete the tutorial **Print the Certificate** stating that you completed the tutorial and email to TA for credit.

COMPLETING THE COURSE:

After studying the material, you must pass the online exam to successfully complete the Safety Certificate. Again, go online to the 'Academy' and Chemical Reactivity Hazards in 'My Stuff' and click "Resume". Click on the Quiz. You may need to disable the pop-up blocker on your browser to allow the link to load. Take the quiz. Once completed you can go back to the main menu in the tutorial and click on "Print Safety Certificate". There is an evaluation as well that you can fill out if you desire.

Note: Once you start the quiz in the popup window, you must navigate away from the page in the main browser or you will not get credit and you will have to take it over again.

GRADING: You must email proof (Certificate issued by SACHE) that you have completed the tutorial and answered the questions for full credit. You can print your Certificate by going to TRAINING HISTORY under "LEARNING". It is in the right hand or left hand menu on the tutorial page.

Regards,

DUE: Thursday, March 30, 2017

Safety Assignment: Industrial Hazards

Laboratory safety is an essential part of the Chemical Engineering Laboratory.

Safety Assignment I :

- 1) Read the selected chapters from Guidelines on Hazardous Chemical Safety from JT Baker Safety Course. (see pdf file)
Chap 3: Toxic chemicals
Chap 8: Flammable chemicals
Chap 9: Corrosive chemicals
Chap 11: Reactive chemicals
Chap 12: Insidious hazards
Chap 13: Compressed gases
- 2) Complete the safety quiz.

DUE: January 19th, beginning of lab period.

SAFETY QUIZ: CHE382-Chemical Engineering Laboratory II

Please answer the following questions testing your knowledge from 6 selected chapters of the JT Baker Hazardous Chemical Safety document; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards and Compressed Gases. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words. (5 points per question)

1. What is the general definition of a toxic chemical?

Any chemical which, when ingested, inhaled, absorbed, injected into or developed within the body, in relatively small amounts, by its chemical action, may cause damage to structure or disturbance to function.

2. Name the general types of damage that a toxic chemical can cause:

Local toxicity, systemic toxicity, acute toxicity, chronic toxicity

3. Name two parameters that can effect toxicity:

Two from list on page 3-3: Mode of entry, physical condition, dose, duration, sensitivity, combined effects, stress, others (must list)

4. What is the DOT (Department of Transportation) definition of a nonflammable compressed gas?

A nonflammable material having a pressure greater than 40 psia at 70°F or 104 psia at 130°F.

5. a) How does one determine how much material is remaining in a non-liquified gas cylinder?
Read the cylinder pressure gauge reading which is proportional to the amount of gas remaining inside the cylinder

b) On page 13-10, which pressure gauge on the two-stage regulator will give you the cylinder pressure reading?

The pressure gauge on the left with pressure readings from 500-3000 psi

6. Other than a single compressed gas, what else can be stored inside a compressed gas cylinder?

Liquefied gas, a gas dissolved in a liquid or solid, or a compressed gas mixture

7. Why do insidious hazards represent a problem to workers?

They represent a problem because the workers may not be aware that a problem exists until chronic, systemic poisoning has occurred.

8. What are the four general categories of insidious hazards?

Unanticipated chemical reactions, unanticipated toxicity, improper facility design, improper handling of equipment and chemicals.

9. Name 2 control measures that can be implemented to reduce insidious hazards in the laboratory.

See list of 8 control measures on page 12-13

10. What is the definition of a reactive chemical?

Reactive chemicals are substances which enter into violent reactions during which the spontaneous liberation of heat and/or gases is too rapid to be safely dissipated by the surroundings.

11. Name the two groups and their sub-headings used to categorize reactive chemicals:

Unstable compounds (Explosives, others)

Unstable mixtures (Explosives, water-reactive compounds, air-sensitive compounds, other mixtures of oxidizing agents with reducing agents)

12. List two precautionary measures for handling and/or storing reactive chemicals.

See list on page 11-8 or short paragraph on 11-7.

13. How are corrosive chemicals generally defined?

Corrosive chemicals are substances that, by direct chemical action, are injurious to body tissue or corrosive to metal.

14. How do corrosive chemicals act on body tissues?

They act through direct contact with skin or eyes, inhalation or ingestion, and by direct chemical reaction, dissolution of essential components, destruction of protein or disruption of cell membranes.

15. True or False: Concentrated alkaline solutions have a more corrosive effect on tissue than most strong inorganic acids.

True

(Strong alkalis gelatinize tissue, forming soluble compounds and thereby produce deep painful, penetrating burns)

16. Name the six categories of common corrosive liquids.

Inorganic acids, organic acids, other inorganics, caustic solutions, organic solvents, other organics.

17. What category of corrosive solids presents the greatest hazard because of their frequent use in industry?

Caustic alkalies

18. Name the three components that make up the FIRE TRIANGLE:

Fuel, oxidizer, ignition source (or energy)

19. What category of flammables is the most common source of fires in industry?

Flammable solvents

20. List two common safety problems and two safety control measures for flammable solvents.

See lists on page 8-11

BONUS QUESTION:

Where is the eyeglass cleaning station located in the Unit Operations Laboratory (Room 120)?